Test Plan

Detecting AI-Generated Text Using ML and NLP Techniques

GitHub: https://github.com/SabirShah1402/Detecting-AI-Generated-Text-Project

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5.1 Scope

Component	In Scope	Out of Scope
Data Engineering	Schema validation, merge logic, null handling, split ratios	Source website uptime
Feature Engineering	TF-IDF output shape, stylometric function correctness	Semantic correctness of GloVe vectors
Classical ML	Training pipeline, prediction output, metric computation	scikit-learn internal implementation
Hybrid CNN	Architecture correctness, forward pass, loss convergence, output shape	GPU driver / CUDA version compatibility
RoBERTa Fine-tuning	Tokenisation, training loop, checkpoint saving, inference speed	HuggingFace library internals
Streamlit App	Input handling, output display, error states, response time	Full cross-browser compatibility testing
Cross-dataset Eval	Train/test swap logic, correct CSV loading, metric computation	Statistical significance testing of differences

5.2 Coding Standards

- **Language and style:** Python 3.10+. PEP 8 enforced via flake8 on every pull request to dev (CI check). Maximum line length: 100 characters.
- **Naming conventions:** Functions and variables in snake_case (e.g., load_hc3_dataset, compute_macro_f1). Classes in CamelCase (e.g., HybridCNNClassifier). Constants in UPPER_SNAKE_CASE (e.g., RANDOM_SEED = 42, MAX_FEATURES = 50000).
- **Docstrings:** Every function includes a Google-format docstring documenting parameters, return type, and raised exceptions. Notebooks contain markdown cells explaining each section's purpose, inputs, and outputs before the code.
- **Error handling:** All file I/O, API calls, and model loading wrapped in try/except blocks with descriptive error messages. Python logging module used at INFO level in scripts — print() is not used for production logging.
- **Reproducibility:** RANDOM_SEED = 42 set globally at the top of every notebook and script. All package versions pinned in requirements.txt. Every training run logs timestamp, hyperparameters, and val metrics to results_log.csv.
- **Commit policy:** No direct pushes to dev or main. All changes via pull request. Commit messages follow: "[T-ID] Brief description" e.g., "[T5] Add dual-stream CNN architecture and GloVe loading".

5.3 Data Quality Testing

- **Schema assertion:** After merge: assert set(df.columns) >= {"text","label","source","dataset"}. Assert df["label"].isin([0,1]).all().
- **Null check:** assert df[["text","label"]].isnull().sum().sum() == 0 after all cleaning steps have run.
- **Minimum length:** assert (df["word_count"] >= 10).all() after the length-filter step.
- **Split integrity:** assert len(train) + len(val) + len(test) == len(full_df). Assert set(train["text"]) ∩ set(test["text"]) == empty (no leakage between splits).
- **Label balance:** Assert each split contains both label values and neither class falls below 10% representation (stratification verification).

5.4 Unit Testing

Test file: tests/test_pipeline.py. All tests run via pytest. CI enforces all tests must pass before any merge to dev.

- **test_flatten_hc3():** Verifies HC3 flattening on a 3-row mock input produces correct row count and correct label assignments for human_answers vs chatgpt_answers.
- **test_tfidf_output_shape():** Verifies TfidfVectorizer output matrix has correct shape (n_samples × max_features) for known dummy input.
- **test_stylometric_functions():** Verifies avg_sentence_length() and punctuation_density() return expected float values for three known test strings.
- **test_cnn_forward_pass():** Verifies Hybrid CNN returns a tensor of shape (batch_size, 1) with values in [0, 1] for a mock input batch.
- **test_app_predict():** Verifies the Streamlit prediction function returns a dict with keys "label" and "confidence" for valid string input >= 10 words.
- **test_split_no_overlap():** Verifies no text string appears in more than one CSV split after the data engineering stage.

5.5 Model Performance Evaluation

- **Primary metric:** Macro F1-score on the fixed test set. Must meet minimum KPI of >= 0.83 for all three models; target >= 0.90 for the best model.
- **Unified comparison:** A single table in the report comparing Accuracy, Macro F1, ROC-AUC, and Precision/Recall for all three models.
- **Dummy baseline:** A majority-class DummyClassifier from scikit-learn defines the floor. Every model must strictly outperform it — failure raises a GitHub Issue before results are reported.
- **Cross-dataset generalisation:** Each model trained on HC3 only → evaluated on Kaggle test set (and vice versa). Results in cross_dataset_results.csv, discussed in report.

5.6 User Acceptance Testing (UAT)

ID	Scenario	Expected Result	Pass Criteria
UAT-01	Paste known human text (Wikipedia paragraph)	Label: Human, confidence >= 60%	Correct label returned
UAT-02	Paste known AI text (ChatGPT output)	Label: AI-Generated, confidence >= 60%	Correct label returned
UAT-03	Submit empty text field	Validation warning shown, app does not crash	Error message displayed
UAT-04	Submit text under 10 words	"Text too short for reliable detection" warning	Warning shown, no crash
UAT-05	Submit 1,000+ word text	Result returned within 3 seconds	Meets response time KPI
UAT-06	Paste text with special characters / symbols	App processes without unhandled exception	No crash
UAT-07	Load app on mobile-width screen (< 480px)	Core layout readable, submit button functional	Usable on mobile browser

UAT is conducted by Person 1 and Person 2 (not involved in app development) using Chrome on desktop and a mobile browser. All results recorded in docs/uat_results.csv. Any failure creates a GitHub Issue assigned to Person 5 or 6, to be resolved before final submission.

5.7 Code Review and Sign-Off

- **Pull request policy:** Every feature branch requires a PR reviewed by at least one other team member. Review checks: PEP 8, docstrings present, unit tests pass, logic correctness.
- **Merge to main:** Person 6 controls merges to main. Requires all CI tests green and two approvals at milestone merges (end Wk 2, Wk 4, Wk 6).
- **Final sign-off:** Tag v1.0 on main after all six members confirm their sections complete. Sign-off entries committed to CHANGELOG.md in the repository root.